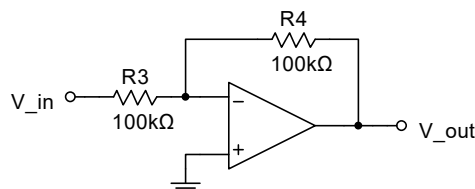


MB Problems: Two Important Kinds of ICs

Answers to MB problems are on page M-12. **Updated Sunday March 1, 2026 3:50pm.**

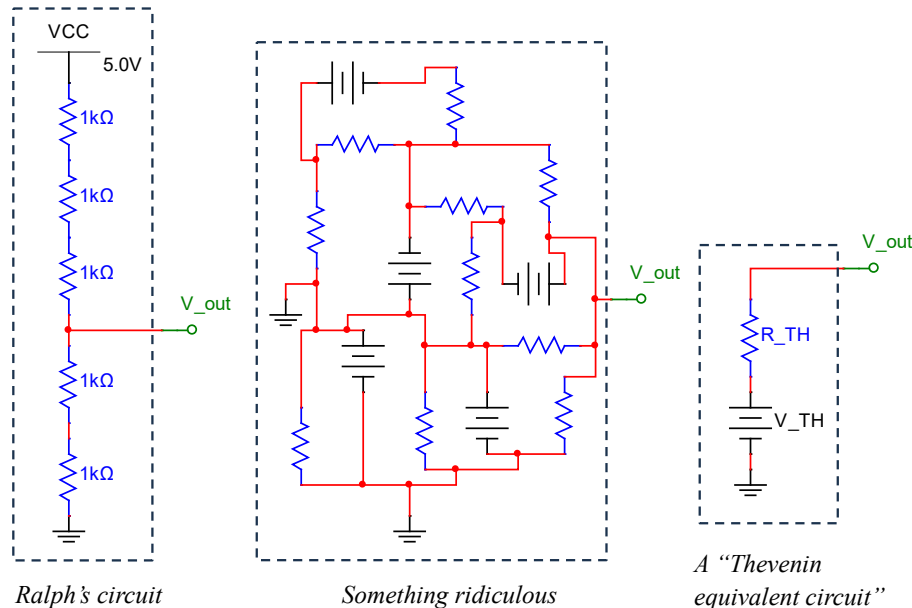
MB1. For the circuit shown below, suppose that $V_{in} = 3$ volts. Find (a) V_+ and V_- , (b) I_{R3} and I_{R4} , (c) ΔV_{R3} and ΔV_{R4} . (d) What direction does current flow in R_3 and in R_4 : to the left or to the right? (e) Which direction does current flow through the op-amp’s output? (f) What is V_{out} ?



MB2. A student named Ralph has used a voltage divider to build a 2-volt DC voltage supply, shown below on the left. It turns out there is a very useful theorem by a fellow named Thevenin, which says roughly the following:

“Any collection of voltage sources and resistors (even the crazy middle figure below!) is functionally equivalent to a single voltage source V_{Th} and a single internal resistance R_{Th} as shown on the right. No measurement at V_{out} can distinguish between the two circuits.” —Léon Charles Thévenin (heavily paraphrased).

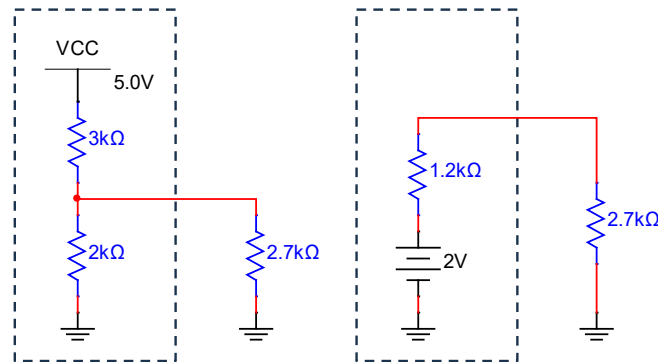
In this problem, you will calculate the values of V_{Th} and R_{Th} for the equivalent circuit of Ralph’s voltage supply.



- What is the open-circuit voltage of the Thevenin equivalent circuit on the right, in terms of V_{Th} and R_{Th} ?
- What is the short-circuit current of the Thevenin equivalent circuit on the right?
- What is the open-circuit voltage of Ralph’s circuit?
- What is the short-circuit current of Ralph’s circuit?
- From your previous parts what are the values of V_{Th} and R_{Th} for Ralph’s circuit?
- Ralph does the same calculation that you did, and says, “Hey, it looks like the output impedance of my circuit is 1.2 kΩ.” Is *output impedance* the right thing for Ralph to call it?

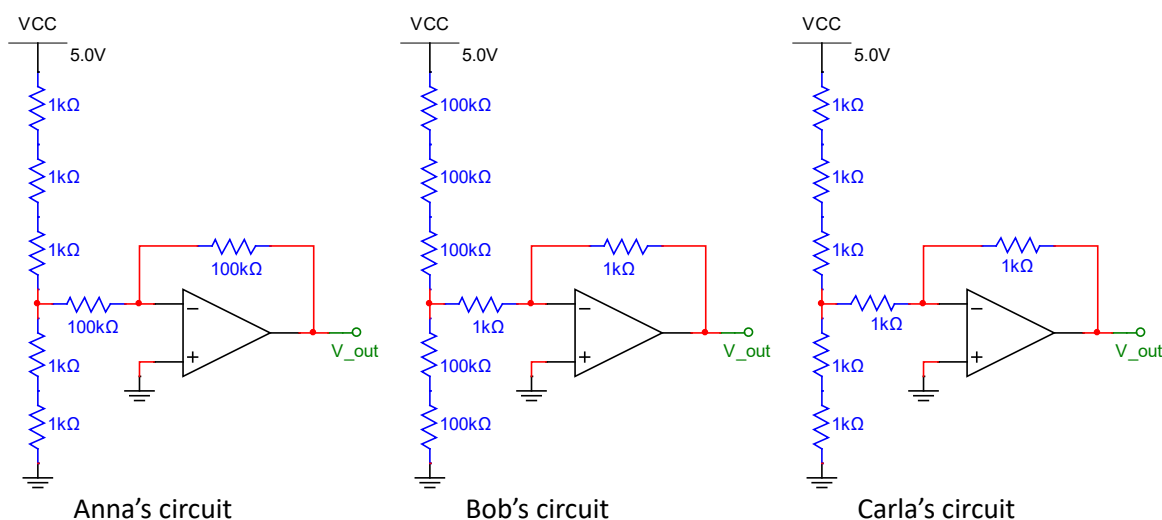
For more help using Thevenin’s theorem, see Appendix G.

MB3. In this problem, you will test whether Ralph’s circuit in the previous problem is equivalent to a Thevenin equivalent circuit with $V_{Th} = 2$ volts and $R_{Th} = 1.2\text{ k}\Omega$, by checking the behavior of a circuit with a single, arbitrarily chosen load resistor R_L . Suppose a load $R_L = 2.7\text{ k}\Omega$ is connected between Ralph’s output and ground. (a) Calculate the voltage across and current through R_L by analyzing the circuit in the diagram below on the left. (b) Calculate the voltage and current for R_L by analyzing the circuit in the diagram below on the right. (c) Are they the same? (d) Does this count as a “proof” of Thevenin’s theorem?



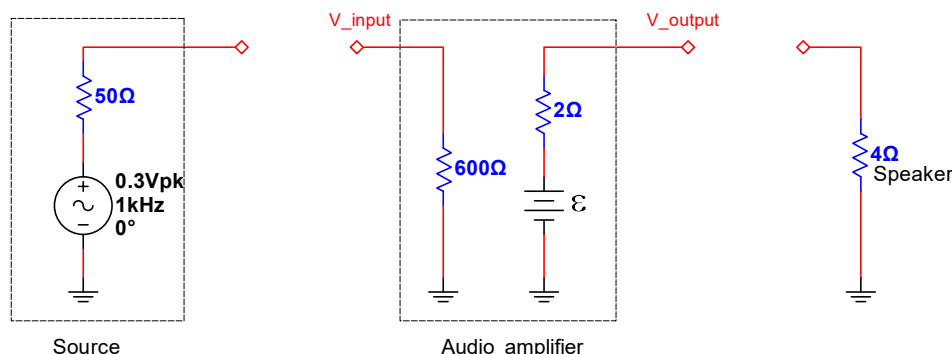
MB4. (a) What is the input impedance of the inverting amplifier you built in Lab 7, part 2? (b) By contrast, is the input impedance of the non-inverting amplifier you built in Lab 7 part 1 larger or smaller than the inverting amplifier? (c) What additional information would you need in order to calculate a numerical value for the input impedance of the non-inverting amplifier, and where would you go to find it?

MB5. Anna and Bob are building inverting amplifiers, just as you did in Lab 7, part 2. Their goal is to use a 2-volt signal as an input to an inverting amplifier with a gain of -1 . Anna builds the circuit on the left (below), and Bob builds the circuit in the middle. (a) Which of the two circuits works the way it’s supposed to, and why does the other one *not* work? (b) Carla is out of $100\text{ k}\Omega$ resistors, so she builds the circuit on the right. Show how to calculate the output voltage of Carla’s circuit *specifically by using the results from problems MB2 and MB4*.

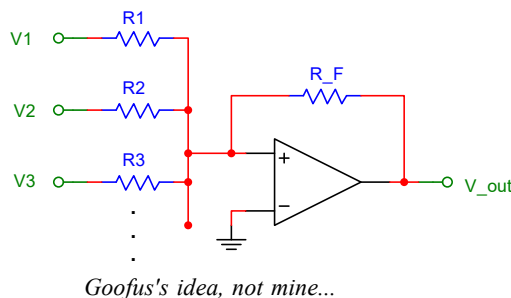


MB6. Referring to the circuit diagrams in parts 1 and 2 of Lab 7, *derive* (using a judicious mix of math, words, and/or pictures) expressions for (a) the gain G of a non-inverting amplifier in terms of R_1 and R_2 , and (b) the gain G of an inverting amplifier in terms of R_3 and R_4 .

MB7. A source with an output impedance of 50 ohms produces an open-circuit sinusoidal voltage with an amplitude of 300 mV. The source is connected to an audio amplifier with voltage gain $G = 200$, input impedance $R_{\text{in}} = 600 \Omega$, and output impedance $R_{\text{out}} = 2 \Omega$. The output of the amplifier is connected to a speaker with an impedance of 4Ω . (You don’t need to worry about the inner workings of the amplifier other than that it produces an internal source emf ε that is 200 times larger than the voltage across its internal 600Ω resistance; that is, $\varepsilon = 100 \times V_{\text{input}}$. (a) Find the amplitude of the voltage across the speaker. (b) Suppose the audio amplifier has a maximum current output of 30 A. What is the maximum amplitude of the input voltage to the amplifier, above which the sinusoidal output to the speaker would be “clipped” as you saw in Lab 7 part 9?



MB8. Goofus and Gallant have been learning about op-amps. Goofus believes that by flipping the V_+ and V_- terminals of the op-amp, he has invented a handy way to build a non-inverting summing amplifier, shown below:



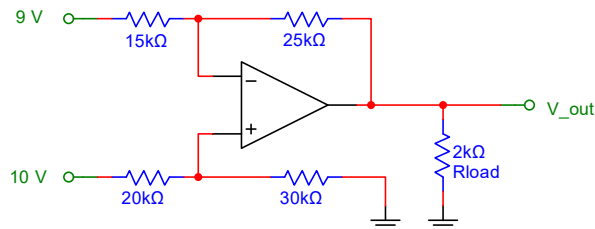
Goofus explains to Gallant that it should work perfectly, according to the golden rules:

- By Golden Rule 2, $V_+ = V_- = 0$ volts.
- By Golden Rule 1, no current flows into the V_+ input, so $I_{RF} = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}$.
- Based on the current through R_F , the output should be $V_{\text{out}} = +R_F \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right)$.

- Write a short, condescending email from Gallant to Goofus that explains what’s wrong with his reasoning.
- Include in your email what the output of his circuit would *actually* be were he to build it, *and why*.

MB9. A noninverting amplifier with a gain $G = 5$ (as in Lab 7 part 2) needs to provide an output of 30 mA to a load at $V_{\text{out}} = 5$ volts. (That is, the load resistance is $R_{\text{load}} = 167 \Omega$.) The maximum output current of the op-amp is 40 mA. (a) What are the minimum values of the two resistors R_1 and R_2 that could be used, so that they don’t draw too much current? (b) Why might very large values for the two resistors, like in the range of several $M\Omega$ or more, also cause your circuit to not work correctly? Note: to avoid these two problems, your default should be to pick R_1 and R_2 (the “feedback resistors”) in the 10 k Ω or 100 k Ω ranges. If you do that, you don’t have to worry about it or calculate anything; it’ll mostly just work.

MB10. The circuit below is essentially a differential amplifier, but with unusual values of all four resistors. It’s not obvious what the use of such a circuit would be, but set that aside and use the “golden rules” of op-amps to calculate (a) the voltages at the inputs V_+ and V_- , (b) the output voltage V_{out} , and (c) the output current of the op-amp. (That is, the current coming out of the output pin of the op-amp, not necessarily the current in the load resistor.)



MB11. On page 749 of our textbook by Paynter, the author relates an op-amp’s maximum usable frequency to its slew rate by plunking down Equation 22.5,

$$f_{\text{max}} = \frac{\text{slew rate}}{2\pi V_{\text{pk}}},$$

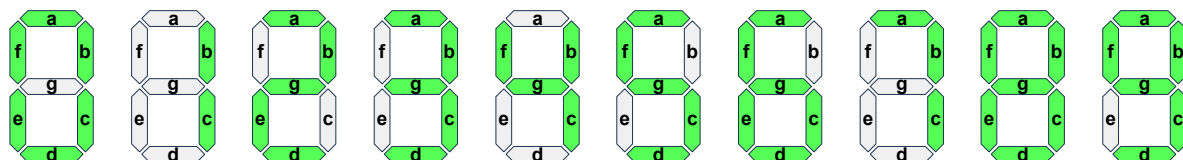
with no explanation of where it might come from or why it might be true. Derive this equation, providing the explanation that Paynter left out. Incidentally, the lack of motivation for equations like this is one of the reasons I’m not crazy about this book. I suspect it stems in part from the assumption that the readers’ math ability is so exceptionally low that they are just *barely* able to plug numbers into a formula that’s provided for them, laboriously pressing the calculator buttons with their knuckle pads and then requiring a long rest.

MB12. Design a circuit using only two op-amps that gives an output voltage V_{out} based on three input voltages V_1 , V_2 and V_3 such that

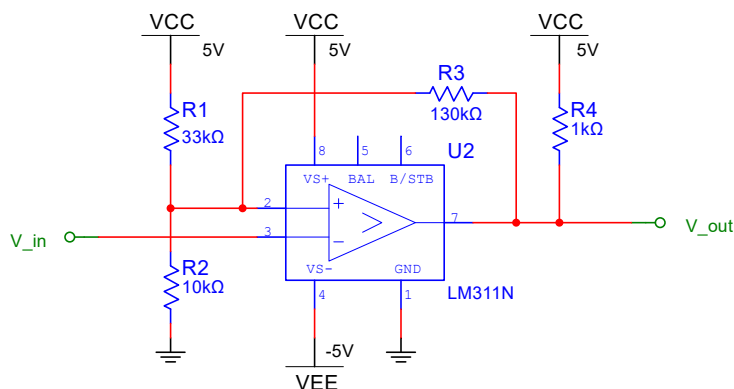
$$V_{\text{out}} = -\left(\frac{V_1}{6} + \frac{V_2 - V_3}{3}\right).$$

MB13. Show how you can create an XOR gate using only a combination of NOR and NAND gates.

MB14. The numbers 0 through 9 are often displayed on clocks or instruments using a familiar *seven segment display* shown below, where the seven individual rectangular LEDs are turned on or off to represent each number. Design a logic circuit to control just one of the LEDs shown below: the one labeled “b” on the upper right side. Your circuit will have four digital inputs representing the 1’s, 2’s, 4’s, and 8’s places of a binary number. (Binary numbers are explained in Mini-Project 2 on page 33.) For this problem, you can assume that these inputs will only represent the numbers 0 through 9 (0000 through 1001), and never anything higher.

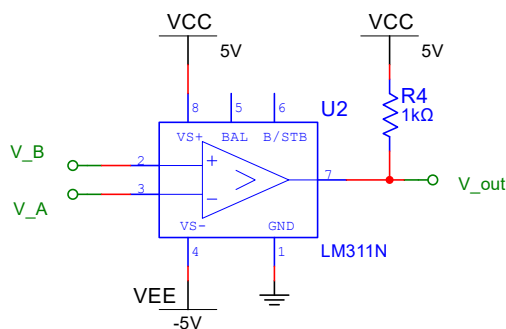


MB15. For the circuit below, calculate the values of V_{in} at which the output switches from high to low and from low to high. You can make the approximation that $R_4 \approx 0$.

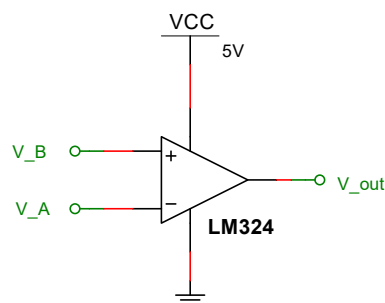


MB16. Design a Schmitt trigger using the LM311 comparator that switches the output from low to high at $V_{in} < 2.6$ volts, and from high to low at $V_{in} > 2.4$ volts.

MB17. Goofus and Gallant are studying comparators, and they both understand that a comparator like the LM311 is really just a slightly specialized op-amp. They both want to build a circuit that will output +5 V or 0 V depending on the relative sizes of the two input voltages. Gallant builds the circuit on the left using an LM311 comparator. Goofus builds the circuit on the right using an LM324 op-amp.



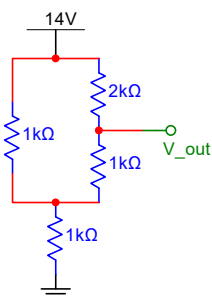
Gallant's comparator



Goofus's comparator

Unfortunately, Goofus did not read the datasheet for the LM324 before building his circuit. (a) If $V_B > V_A$, does Goofus's output really equal 5 V, bearing in mind that the correct answer is “No”? Based on the LM324's datasheet, what will its actual high output be? (b) Based on the datasheets for the LM311 and the LM324, roughly how long will it take each of their circuits to swing from low output to high output? (The datasheet I gave you for the LM324 doesn't list a specific slew rate, but there's a graph on page 6 that gives a rough idea of it.)

MB18. Draw a Thevenin equivalent circuit for the circuit below, including calculating values for V_{Th} and R_{Th} .



MB19. Examine *at least two* other students’ lab notebooks. (Students who would prefer to not have their notebooks looked at can let me know and leave your notebooks on my table at the front of the room for a few days.) Write me a paragraph or so about one or more things you notice about these notebooks that suggest something that you could do differently to improve your own lab notebook. Its up to you whether to name the other students. If you truly believe that your notebook is superior in all aspects to all other notebooks, then you can write about that instead—but be prepared for me to gently rebut your claim.

Answers to Selected MB Problems:

MB1. (a) 0 (b) $30\ \mu\text{A}$ (c) 3 volts (d) both flow left to right (e) right to left (f) -3 volts

MB2. (a) It’s just V_{Th} (b) V_{Th} and R_{Th} (c) 2 V (d) 1.67 mA (e) 2 V and 1.2 k Ω (f) Sure, that’s basically what it is. In this context, “output impedance/resistance” or “internal resistance” or “Thevenin equivalent resistance” basically all mean the same thing.

MB3. (a) 1.38 V and 513 μA (c) They’d better both be the same! (d) No, it’s not a general proof, because you demonstrated it only for one specific load (a single 2.7 k Ω resistor) connected to one specific circuit made by some fake guy named Ralph. But if you see this kind of thing enough times, it does start to seem convincing. :-)

MB4. (a) Hint: The input V_+ is at ground, and the op-amp adjusts its output to keep $V_- = V_+$. This means that the V_- input is essentially at 0 volts as well—what we call a *pseudo ground*—and you can treat it as if the right side of resistor R_3 was simply connected directly to ground.

MB5. (b) -0.91 volts

MB6. (a) $G = (R_1 + R_2)/R_2$ (b) $G = -R_4/R_3$. General hint: In problem MB1 you worked out the gain of an inverting amplifier with specific resistor values, and you can follow the same kind of logic here.

MB7. (a) 37 V (b) 900 mV

MB8. Hints: (a) Your explanation to Goofus will almost surely involve the words *feedback*, *negative*, and *positive*. (b) The actual output is guaranteed to be either $\approx +15$ V or ≈ -15 V, pinned at one of the two power rails. But Goofus won’t intuit this on his own, so you still need to explain to him why.

MB9. (a) 100 Ω and 400 Ω (b) Hint: what might be a realistic value for the input impedance of the V_- input?

MB10. (a) $V_+ = V_- = 6$ volts (b) 1 volt (c) 300 μA

MB15. 1.38 V and 1.10 V

MB16. Using the diagram in problem MB15, set $R_1 = R_2 = R_3/12$

MB17. (b) Gallant’s circuit is more than ten times faster than Goofus’s.

MB18. $V_{\text{Th}} = 10$ V, $R_{\text{Th}} = 892\ \Omega$.